

ACADEMIC ASSIGNMENT

Computer Applications & Financial Literacy

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PART 1: COMPUTER SCIENCE — SPREADSHEET FUNCTIONS

In modern spreadsheet software such as Microsoft Excel, Google Sheets, and LibreOffice Calc, **functions** are predefined formulas designed to perform complex operations, calculations, and data processing automatically. Every function begins with an equals sign (=) followed by the function name and parameters/arguments enclosed in parentheses.

Table 1: Mathematical Functions

Mathematical functions perform numerical calculations, statistical aggregations, and algebraic transformations across selected ranges of cells.

Function	Description & Purpose	Syntax Example	Input Data	Output
SUM	Calculates the total aggregate sum of all numerical values in a specified cell range.	=SUM(A1:A3)	15, 25, 10	50
AVERAGE	Computes the arithmetic mean of a set of numbers by dividing total sum by count.	=AVERAGE(B1:B3)	10, 20, 30	20
COUNT	Counts the total number of cells containing numerical data within a given range.	=COUNT(C1:C4)	5, "Hi", 12, 8	3
MAX	Scans a numerical dataset and returns the highest (maximum) value.	=MAX(12, 89, 45, 102)	12, 89, 45, 102	102
MIN	Scans a numerical dataset and returns the lowest (minimum) value.	=MIN(12, 89, 45, 102)	12, 89, 45, 102	12
SQRT	Calculates the positive square root of a non-negative real number.	=SQRT(64)	64	8
POWER	Raises a given base number to a specified exponential power.	=POWER(3, 4)	Base: 3, Exp: 4	81
MOD	Calculates and returns the remainder after integer division.	=MOD(17, 5)	Num: 17, Div: 5	2
ROUND	Rounds a floating-point number to a specified number of decimal places.	=ROUND(14.867, 2)	14.867, Dec: 2	14.87
ABS	Returns the absolute (non-negative) value of a number, stripping minus signs.	=ABS(-42.5)	-42.5	42.5

Table 2: Text Functions

Text functions enable manipulation, extraction, formatting, and analysis of alphanumeric text strings within cells.

Function	Description & Purpose	Syntax Example	Input Text	Output
CONCATENATE	Joins multiple individual text strings or cell values into a single text sequence.	=CONCATENATE("Data", " ", "Science")	"Data", "Science"	"Data Science"
LEN	Returns the exact total character length of a string, including spaces and symbols.	=LEN("Spreadsheet")	"Spreadsheet"	11
UPPER	Converts all alphabetical characters within a target string to uppercase letters.	=UPPER("computer")	"computer"	"COMPUTER"
LOWER	Converts all alphabetical characters within a target string to lowercase letters.	=LOWER("NETWORK")	"NETWORK"	"network"
PROPER	Capitalizes the first letter of each word in a string while converting rest to lowercase.	=PROPER("john smith")	"john smith"	"John Smith"
LEFT	Extracts a specified number of characters from the beginning (left) of a string.	=LEFT("Database", 4)	"Database", 4	"Data"
RIGHT	Extracts a specified number of characters from the end (right) of a string.	=RIGHT("Database", 4)	"Database", 4	"base"
MID	Extracts a substring from the middle of a text string, starting at a specific position.	=MID("Programming", 4, 4)	"Programming", 4, 4	"gram"
TRIM	Removes leading, trailing, and duplicate spaces from text, leaving single spaces between words.	=TRIM(" System ")	" System "	"System"
TEXTJOIN	Concatenates a range of text strings using a custom delimiter, with an option to ignore empty cells.	=TEXTJOIN("-", TRUE, "A", "B")	"-", TRUE, "A", "B"	"A-B"

PART 2: FINANCIAL LITERACY — COMPREHENSIVE NOTE ON MINIMUM BALANCE

1. Definition and Overview

A **Minimum Balance** is the minimum threshold dollar or currency amount that a banking customer is contractually required to maintain in their bank account (such as a savings or checking account) to keep the account active without incurring operational charges. Financial institutions institute minimum balance guidelines to ensure they retain sufficient liquid reserves and cover maintenance costs associated with account servicing, electronic infrastructure, and record-keeping.

2. Primary Methods of Calculation

Banks calculate minimum balance requirements using two standard banking conventions:

- **Minimum Daily Balance Method:** Under this strict standard, the account balance must never fall below the stipulated minimum threshold at any point during any given day of the billing cycle. If the balance drops below this threshold even for a single day, a non-maintenance penalty fee is triggered.
- **Average Monthly Balance (AMB) Method:** This approach offers greater flexibility. The bank calculates the sum of the daily closing balances for every day in the calendar month and divides this total by the number of days in that month.

$$\text{Average Monthly Balance (AMB)} = \frac{\sum \text{Daily Closing Balances}}{\text{Total Number of Days in the Month}}$$

3. Financial Implications of Non-Maintenance

Failing to maintain the required balance can lead to several financial and operational consequences:

1. **Non-Maintenance Penalties:** Banks automatically deduct penalty charges directly from the account balance, which can cause the balance to decrease even further.
2. **Depletion of Base Capital:** Continuous monthly penalties can gradually erode the account holder's primary savings over time.
3. **Account Freezing or Status Restrictions:** Persistent failure to maintain the threshold can result in temporary account freezing, loss of debit card or online banking features, and potential account closure.

Exceptions & Zero-Balance Accounts:

To promote financial inclusion, many banks offer *Zero-Balance Accounts* specifically tailored for students, basic salary earners, and government assistance programs. These accounts carry no minimum balance requirement and are completely exempt from non-maintenance penalty fees.